

Chapter 6 – Capacitance

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
3	Formulate the capacitance with one dimensional potential variation using direct integration	A2, F	C5	2

Objectives:

- 1. Definition of Capacitance
- 2. Resistors and Capacitors
- 3. Methods for Calculating Capacitance for symmetric structures using different methods.
 - a) Method 1: Find the **E** field and then find *V*.
 - b) Method 2: Finding *C* using one-dimensional change in potential.

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Capacitance

Capacitance of two conductor systems as the ratio of the magnitude of the total charge on either conductor to the potential difference between conductors

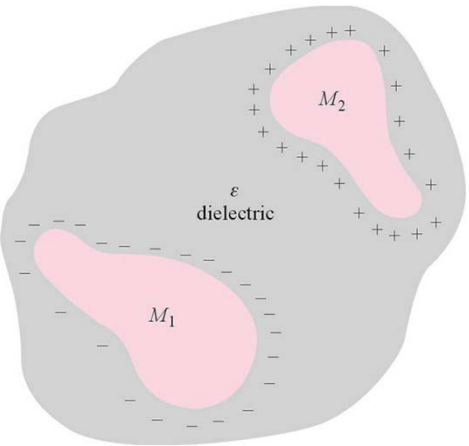
$$C = \frac{Q}{V_0}$$

- *Q* is determined by a surface integral over the positive conductors
- *V*₀ is found by carrying a unit positive charge from the negative to the positive surface.

$$C = \frac{Q}{V_0} = \frac{\oint_S \epsilon \mathbf{E} \cdot d\mathbf{S}}{-\int_-^+ \mathbf{E} \cdot d\mathbf{L}}$$

Capacitance is independent of the potential and the total charge. It's a function only of the physical dimensions of the system of conductors and of the permittivity of the homogeneous dielectric.
Capacitance is measured in Farads (F) – one Coulomb per Volt.

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Parallel-Plate Capacitor

Find the capacitance of infinite parallel planes with separation distance d .

Conductor surface

ϵ

Uniform surface charge density

Conductor surface

$-\rho_s$

$z = d$

$\mathbf{n}_u$

$\mathbf{n}_l$

$z = 0$

$\mathbf{E}$

$C = \frac{Q}{V_0} = \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{-\int_-^+ \mathbf{E} \cdot d\mathbf{L}}$

METHOD 1: Field due to Surface Charge

$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_-$
 $\mathbf{E} = \frac{\rho_s}{2\epsilon} \mathbf{a}_z + \frac{\rho_s}{2\epsilon} \mathbf{a}_z = \frac{\rho_s}{\epsilon} \mathbf{a}_z$
 $\mathbf{D} = \epsilon \mathbf{E} = \rho_s \mathbf{a}_z$

METHOD 2: Using Boundary Conditions at any one of the plate surfaces

Lower Plate

$\mathbf{D} \cdot \mathbf{n}_l \Big|_{z=0} = \mathbf{D} \cdot \mathbf{a}_z = \rho_s$
 $\mathbf{D} = \rho_s \mathbf{a}_z$

Upper Plate

$\mathbf{D} \cdot \mathbf{n}_u \Big|_{z=d} = \mathbf{D} \cdot (-\mathbf{a}_z) = -\rho_s$
 $\mathbf{D} = \rho_s \mathbf{a}_z$

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Parallel-Plate Capacitor

Find the capacitance of infinite parallel planes with separation distance d .

Conductor surface

ϵ

Uniform surface charge density

Conductor surface

$-\rho_s$

$z = d$

$\mathbf{n}_u$

$\mathbf{n}_l$

$z = 0$

$\mathbf{E}$

$C = \frac{Q}{V_0} = \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{-\int_-^+ \mathbf{E} \cdot d\mathbf{L}}$

$\mathbf{E} = \frac{\rho_s}{\epsilon} \mathbf{a}_z$
 $\mathbf{D} = \epsilon \mathbf{E} = \rho_s \mathbf{a}_z$

$Q = \rho_s S$
 $V_0 = \frac{\rho_s}{\epsilon} d$

The potential difference between lower and upper planes is,

$V_0 = -\int_{upper}^{lower} \mathbf{E} \cdot d\mathbf{L} = -\int_{z=d}^0 \frac{\rho_s}{\epsilon} dz = \frac{\rho_s}{\epsilon} d$

The energy stored in the capacitor is,

$W_E = \frac{1}{2} \int_{vol} \epsilon E^2 dv = \frac{1}{2} \int_0^S \int_{z=0}^d \epsilon \frac{\rho_s^2}{\epsilon^2} dz dS = \frac{1}{2} \frac{\rho_s^2}{\epsilon} S d = \frac{1}{2} \frac{\epsilon S \rho_s^2}{\epsilon^2} d^2$

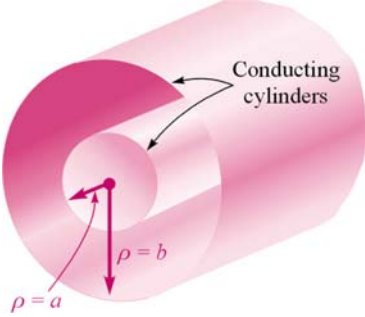
$C = \frac{Q}{V_0} = \frac{\epsilon S}{d}$
 $W_E = \frac{1}{2} Q V_0 = \frac{1}{2} \frac{Q^2}{C}$
 $W_E = \frac{1}{2} C V_0^2$

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Several Capacitance Examples

Coaxial Cable



Conducting cylinders

$\rho = a$

$\rho = b$

$$\mathbf{D} = \frac{a\rho_s}{\rho} \mathbf{a}_\rho = \epsilon \mathbf{E} \quad (a < \rho < b)$$

$$\mathbf{E} = \frac{a\rho_s}{\rho\epsilon} \mathbf{a}_\rho \quad (a < \rho < b)$$

$$V_{ab} = - \int_{\rho=b}^a \mathbf{E} \cdot d\mathbf{L} = - \int_b^a \frac{a\rho_s}{\rho\epsilon} d\rho$$
$$V_{ab} = \frac{a\rho_s}{\epsilon} \ln \frac{b}{a}$$

$$C = \frac{Q}{V_{ab}} = \frac{2\pi a\epsilon L \rho_s}{a\rho_s} \ln \frac{b}{a}$$

$$C = \frac{2\pi\epsilon L}{\ln \frac{b}{a}}$$

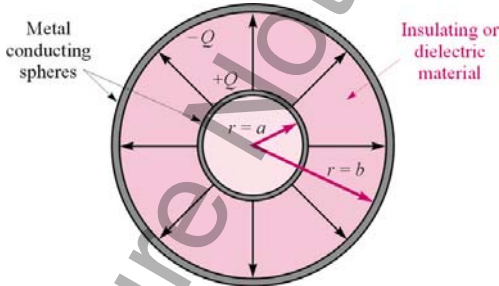
$$Q = \int_S \rho_s dS_\rho = \int_{z=0}^L \int_{\varphi=0}^{2\pi} \rho_s \rho d\varphi dz$$
$$Q = 2\pi a L \rho_s$$

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Several Capacitance Examples

Two Concentric Spheres



Metal conducting spheres

$-Q$

$+Q$

$r = a$

$r = b$

Insulating or dielectric material

$$\mathbf{E} = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \quad (a < r < b)$$

$$V_{ab} = - \int_{r=b}^a \mathbf{E} \cdot d\mathbf{L}$$
$$V_{ab} = - \int_b^a \frac{Q}{4\pi\epsilon r^2} dr$$
$$V_{ab} = \frac{Q}{4\pi\epsilon} \left(\frac{1}{a} - \frac{1}{b} \right)$$

If $b \rightarrow \infty$, Capacitance of an isolated spherical conductor is;

$$C = 4\pi\epsilon a$$

$$C = \frac{Q}{V_{ab}} = \frac{4\pi\epsilon}{\left(\frac{1}{a} - \frac{1}{b} \right)}$$

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Series and Parallel Capacitors

- If two capacitors with capacitance C_1 and C_2 are in series (having same charge on them), the total capacitance is;

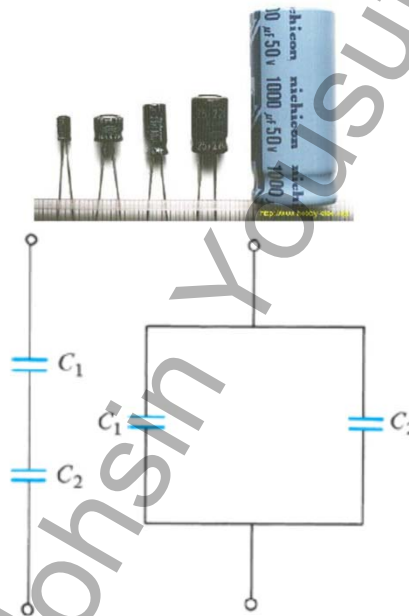
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

or

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

- If the capacitors are in Parallel (having same voltage across them), the total capacitance is:

$$C = C_1 + C_2$$



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Resistors and Capacitors

- It has been shown that,

$$R = \frac{V_{ab}}{I} = \frac{-\int_b^a \mathbf{E} \cdot d\mathbf{L}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{S}}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

and

$$C = \frac{Q}{V_{ab}} = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{S}}{-\int_b^a \mathbf{E} \cdot d\mathbf{L}}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

- The product of these expressions yields,

$$RC = \frac{\epsilon}{\sigma}$$

which is the relaxation time T_r of the medium separating the conductors

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Resistors and Capacitors

- Assuming homogeneous media, the resistance of various capacitors mentioned earlier can be obtained using; $RC = \epsilon/\sigma$

- For a parallel-plate Capacitor,

$$C = \frac{\epsilon S}{d}, \quad R = \frac{d}{\sigma S}$$

- For a Cylindrical Capacitor,

$$C = \frac{2\pi\epsilon L}{\ln \frac{b}{a}}, \quad R = \frac{\ln \frac{b}{a}}{2\pi\sigma L}$$

- For a Spherical Capacitor,

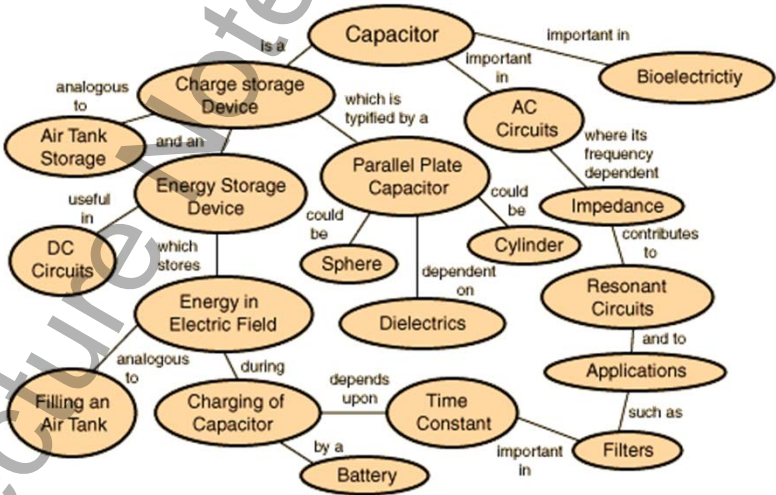
$$C = \frac{4\pi\epsilon}{\left(\frac{1}{a} - \frac{1}{b}\right)}, \quad R = \frac{\left(\frac{1}{a} - \frac{1}{b}\right)}{4\pi\sigma}$$

Note: R in these equations is NOT the resistance of the capacitor plate, but the leakage resistance between the plates.

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Capacitance



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Poisson's and Laplace Equations: 2nd Method of Finding C

From the point form of Gauss's Law,

$$\nabla \cdot \mathbf{D} = \rho_v$$

where,

$$\mathbf{D} = \epsilon \mathbf{E}$$

and the gradient relationship,

$$\mathbf{E} = -\nabla V$$

So,

$$\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon}$$

In RCS,

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

$$\nabla \cdot \nabla V = \frac{\partial}{\partial x} \left(\frac{\partial V}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial V}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial V}{\partial z} \right)$$

$$\nabla \cdot \nabla V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \nabla^2 V$$

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}$$

Poisson's Equation

$$\nabla^2 V = 0$$

Laplace's Equation

If $\rho_v = 0$, indicating zero *volume* charge density, but allowing point charges, line charge, and surface charge density to exist at singular locations as sources of the field, then

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Laplacian Operator in Coordinate Systems

$$\nabla^2 V = 0$$

- The ∇^2 operation is called the *Laplacian* of V .
- Remember that, $\nabla^2 V = \nabla \cdot \nabla V$
- The divergence of the gradient of a scalar function is called the *Laplacian*.
- In RCS,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2} \quad (\text{cylindrical})$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \quad (\text{spherical})$$

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Uniqueness Theorem

Given is a volume v with a closed surface S . The function $V(x, y, z)$ is completely determined on the surface S . There is only one function $V(x, y, z)$ with given values on S (the boundary values) that satisfies the Laplace equation.

Application: The theorem of uniqueness allows to make statements about the potential in a region that is free of charges if the potential on the surface of this region is known. The Laplace equation applies to a region of space that is free of charges. Thus, if a region of space is enclosed by a surface of known potential values, then there is only one possible potential function that satisfies both the Laplace equation and the boundary conditions.

Example: A piece of metal has a fixed potential, for example, $V = 0 \text{ V}$. Consider an empty hole in this piece of metal. On the boundary S of this hole, the value of $V(x, y, z)$ is the potential value of the metal, i.e., $V(S) = 0 \text{ V}$. $V(x, y, z) = 0$ satisfies the Laplace equation (check it!). Because of the theorem of uniqueness, $V(x, y, z) = 0$ describes also the potential inside the hole

Uniqueness Theorem and Boundary Conditions

RCS
 $V(x, y, z)$

CCS
 $V(\rho, \varphi, z)$

SCS
 $V(r, \theta, \varphi)$

For one-dimensional change in potential, i.e. $V = f(\text{one variable})$
How many unique Capacitor structures are formed?
Remember the planes and surfaces formed by each of the variable.
For example, $\rho = \text{const.}$ is a cylinder.

Five unique Capacitor structures are formed

- 1) $V(x), x = \text{const.}$ is a plane, Parallel-Plate Capacitor.
- 2) $V(\rho), \rho = \text{const.}$ is a cylinder, Cylindrical (co-axial) Capacitor.
- 3) $V(\varphi), \varphi = \text{const.}$ is half-plane, Door Capacitor.
- 4) $V(r), r = \text{const.}$ is a sphere, Spherical Capacitor.
- 5) $V(\theta), \theta = \text{const.}$ is a cone, Conical Capacitor.

Examples of the Solution of Laplace's Equation

The method of direct integration is applicable only to problems that are "one-dimensional".

1) $V(x)$, $x = \text{const.}$ is a plane, **Parallel-Plate Capacitor.**

$$\nabla \cdot \nabla V = \nabla^2 V = 0$$

$$\frac{\partial^2 V}{\partial x^2} = 0$$

Since V is a function of x only, so, it becomes a definite derivative,

$$\frac{d^2 V}{dx^2} = 0$$

Integrating twice,

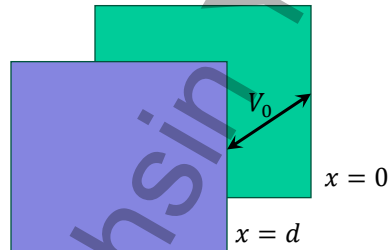
$$\frac{dV}{dx} = A$$

and

$$V = Ax + B, \quad (1)$$

A and B are constants of integration. They are determined from Boundary Conditions only.

Since V is a function of x only, then V is constant if x is a constant, thus, forming equipotential surfaces (parallel planes normal to the x axis).



Boundary Conditions:

Let $V = 0$ at $x = 0$ and $V = V_0$ at $x = d$.

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Examples of the Solution of Laplace's Equation

The method of direct integration is applicable only to problems that are "one-dimensional".

1) $V(x)$, $x = \text{const.}$ is a plane, **Parallel-Plate Capacitor.**

$$V = Ax + B, \quad (1)$$

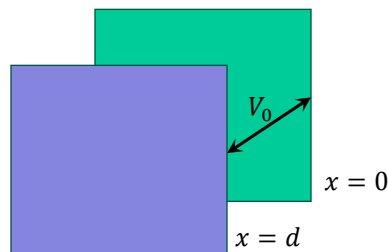
$$A = \frac{V_0}{d}, \quad B = 0$$

$$V = \frac{V_0 x}{d}, \quad (2)$$

Following are the necessary steps to calculate the Capacitance,

1. Given V , find $\mathbf{E}$ using gradient.
2. Use $\mathbf{D} = \epsilon \mathbf{E}$ to find $\mathbf{D}$.
3. Evaluate $\mathbf{D}$ at either capacitor plate, $\mathbf{D} = \mathbf{D}_S = D_N \mathbf{a}_N$.
4. Recognize that $\rho_S = D_N$.
5. Find Q by a surface integration over the capacitor plate, $Q = \int_S \rho_S dS$

Since V is a function of x only, then V is constant if x is a constant, thus, forming equipotential surfaces (parallel planes normal to the x axis).



Boundary Conditions:

Let $V = 0$ at $x = 0$ and $V = V_0$ at $x = d$.

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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

1) $V(x)$, $x = \text{const.}$ is a plane, **Parallel-Plate Capacitor.**

$$\begin{aligned} V &= Ax + B, & (1) \\ A &= \frac{V_0}{d}, & B = 0 \\ V &= \frac{V_0 x}{d}, & (2) \end{aligned}$$

Following are the necessary steps to calculate the Capacitance,

1. Given V , find $\mathbf{E}$ using gradient.
2. Use $\mathbf{D} = \epsilon \mathbf{E}$ to find $\mathbf{D}$.
3. Evaluate D at either capacitor plate, $\mathbf{D} = \mathbf{D}_S = D_N \mathbf{a}_N$.
4. Recognize that $\rho_S = D_N$.
5. Find Q by a surface integration over the capacitor plate, $Q = \int_S \rho_S dS$

Step 1

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x \\ \mathbf{E} &= -\frac{\partial}{\partial x} \left(\frac{V_0 x}{d} \right) \mathbf{a}_x \\ \mathbf{E} &= -\frac{V_0}{d} \mathbf{a}_x \end{aligned}$$

Step 2

$$\mathbf{D} = \epsilon \mathbf{E} = -\epsilon \frac{V_0}{d} \mathbf{a}_x$$

Step 3

$$\begin{aligned} \mathbf{D}_S = \mathbf{D} \Big|_{x=0} &= -\epsilon \frac{V_0}{d} \mathbf{a}_x \\ \mathbf{a}_N &= \mathbf{a}_x, \quad \mathbf{D}_S \cdot \mathbf{a}_x = \rho_S \end{aligned}$$

Step 4

$$D_N = -\epsilon \frac{V_0}{d} = \rho_S$$

Step 5

$$Q = \int_S -\frac{\epsilon V_0}{d} dS = -\frac{\epsilon V_0 S}{d}$$

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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

1) $V(x)$, $x = \text{const.}$ is a plane, **Parallel-Plate Capacitor.**

Step 1

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x \\ \mathbf{E} &= -\frac{\partial}{\partial x} \left(\frac{V_0 x}{d} \right) \mathbf{a}_x \\ \mathbf{E} &= -\frac{V_0}{d} \mathbf{a}_x \end{aligned}$$

Step 2

$$\mathbf{D} = \epsilon \mathbf{E} = -\epsilon \frac{V_0}{d} \mathbf{a}_x$$

Step 3

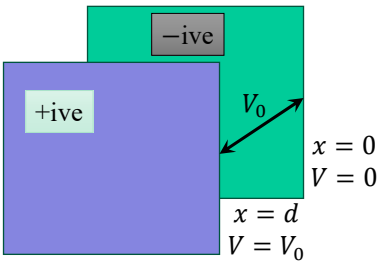
$$\begin{aligned} \mathbf{D}_S = \mathbf{D} \Big|_{x=0} &= -\epsilon \frac{V_0}{d} \mathbf{a}_x \\ \mathbf{a}_N &= \mathbf{a}_x \end{aligned}$$

Step 4

$$D_N = -\epsilon \frac{V_0}{d} = \rho_S$$

Step 5

$$Q = \int_S -\frac{\epsilon V_0}{d} dS = -\frac{\epsilon V_0 S}{d}$$



$$\begin{aligned} C &= \frac{|Q|}{V_0} \\ C &= \frac{\epsilon S}{d} \end{aligned}$$

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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

2. $V(\rho), \rho = \text{const.}$ is a cylinder, Cylindrical (co-axial) Capacitor.

$$\nabla \cdot \nabla V = \nabla^2 V = 0$$
$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0$$

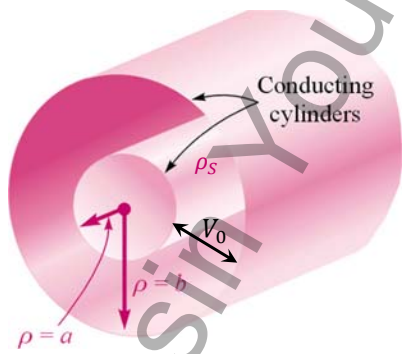
ρ in the denominator, so, $\rho \neq 0$,
multiply by ρ and integrate,

$$\rho \frac{dV}{d\rho} = A$$

Integrating again after rearranging,
We get,

$$V = A \ln \rho + B, \quad (1)$$

The equipotential surfaces are given by
 $\rho = \text{constant}$ and are cylinders, and the
problem is that of the coaxial capacitor
or coaxial transmission line.



Boundary Conditions:
Let $V = 0$ at $\rho = b$ and $V = V_0$ at $\rho = a$.

Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

2. $V(\rho), \rho = \text{const.}$ is a cylinder, Cylindrical (co-axial) Capacitor.

$$V = A \ln \rho + B, \quad (1)$$

Using Boundary Conditions,
 $0 = A \ln b + B$
$$B = -A \ln b = A \ln \frac{1}{b}$$

The second one,
 $V_0 = A \ln a + B$
$$V_0 = A \ln a + A \ln \frac{1}{b}$$
$$V_0 = A (\ln a/b)$$
$$A = \frac{V_0}{\ln a/b}$$
$$B = \frac{V_0 \ln \frac{1}{b}}{\ln a/b}$$

Eq(1) $\Rightarrow$
$$V = \frac{V_0}{\ln a/b} \ln \rho + \frac{V_0}{\ln a/b} \ln \frac{1}{b}$$
$$V = V_0 \frac{\ln \rho/b}{\ln a/b}$$

For $b > a$,
$$V = V_0 \frac{\ln b/\rho}{\ln b/a}, \quad (2)$$

Step 1 $\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{a}_\rho$
$$\mathbf{E} = -\frac{V_0}{\ln b/a} \frac{d}{d\rho} (\ln b - \ln \rho) \mathbf{a}_\rho$$

Boundary Conditions:
Let $V = 0$ at $\rho = b$ and $V = V_0$ at $\rho = a$.

Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

2. $V(\rho), \rho = \text{const.}$ is a cylinder, Cylindrical (co-axial) Capacitor.

Step 1

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{a}_\rho$$
$$\mathbf{E} = -\frac{V_0}{\ln(b/a)} \frac{d}{d\rho} (\ln b - \ln \rho) \mathbf{a}_\rho$$
$$\mathbf{E} = \frac{V_0}{\ln(b/a)} \frac{d}{d\rho} (-\ln b + \ln \rho) \mathbf{a}_\rho$$
$$\mathbf{E} = \frac{V_0/\rho}{\ln(b/a)} \mathbf{a}_\rho = \frac{V_0}{\rho \ln(b/a)} \mathbf{a}_\rho$$

Step 2

$$\mathbf{D} = \epsilon \mathbf{E} = \frac{\epsilon V_0}{\rho \ln(b/a)} \mathbf{a}_\rho$$

Step 3

$$\mathbf{D}_S = \mathbf{D} \Big|_{\rho=a} = \frac{\epsilon V_0}{a \ln(b/a)} \mathbf{a}_\rho$$
$$\mathbf{a}_N = \mathbf{a}_\rho, \quad \mathbf{D}_S \cdot \mathbf{a}_\rho = \rho_S$$

Step 4

$$D_N = \frac{\epsilon V_0}{a \ln(b/a)} = \rho_S$$

Step 5

$$Q = \int_S \frac{\epsilon V_0}{a \ln(b/a)} dS_\rho$$
$$Q = \int_{z=0}^L \int_{\varphi=0}^{2\pi} \frac{\epsilon V_0}{a \ln(b/a)} \rho d\varphi dz$$
$$Q = \frac{2\pi a \epsilon V_0 L}{a \ln(b/a)}$$

Finally,

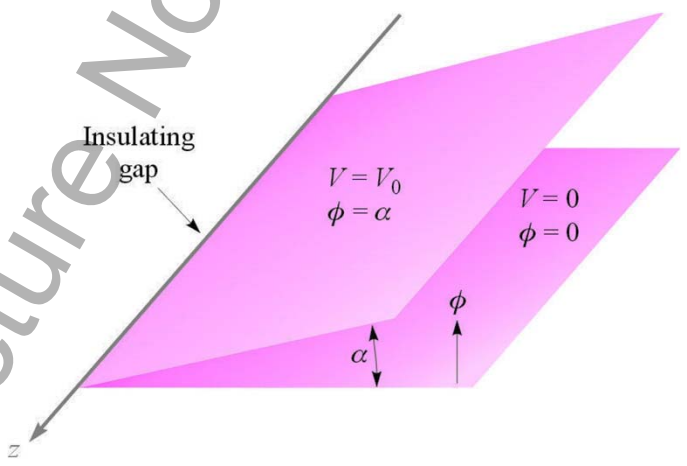
$$C = \frac{|Q|}{V_0}$$
$$C = \frac{2\pi \epsilon L}{\ln(b/a)}$$

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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

3. $V(\phi), \phi = \text{const.}$ is half-plane, Door Capacitor.



The diagram illustrates a 'Door Capacitor' configuration. It shows two infinite parallel plates. The left plate is at potential $V = V_0$ and $\phi = \alpha$. The right plate is at potential $V = 0$ and $\phi = 0$. The region between the plates is shaded pink. An 'Insulating gap' is indicated between the plates. The coordinate ϕ is shown as a vertical arrow, and α is shown as a vertical arrow indicating the position of the left plate. A z -axis is shown pointing downwards from the top-left corner.

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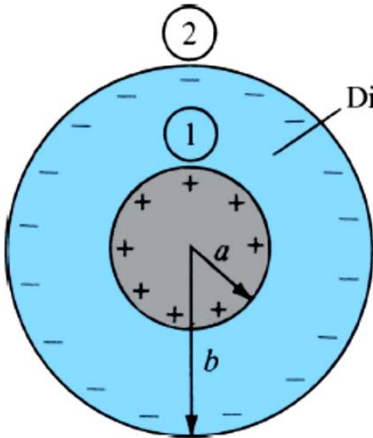
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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

4. $V(r), r = \text{const.}$ is a sphere, **Spherical Capacitor**.



$$V = V_0 \frac{\frac{1}{r} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}$$
$$C = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$$

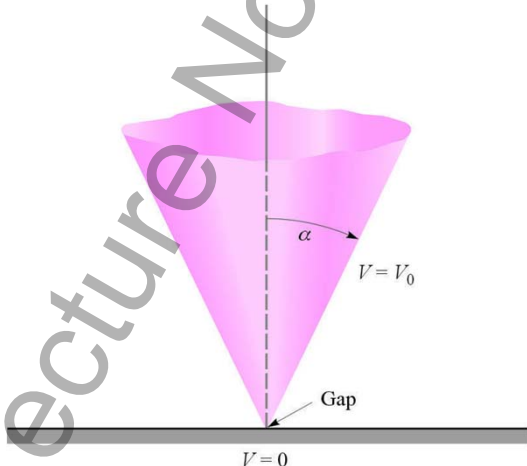
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Examples of the Solution of Laplace’s Equation

The method of direct integration is applicable only to problems that are “one-dimensional”.

5. $V(\theta), \theta = \text{const.}$ is a cone, **Conical Capacitor**.



$$V = V_0 \frac{\ln\left(\tan \frac{\theta}{2}\right)}{\ln\left(\tan \frac{\alpha}{2}\right)}$$
$$C \doteq \frac{2\pi\epsilon r_1}{\ln\left(\cot \frac{\alpha}{2}\right)}$$

D6.6..

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Practice Example # 01

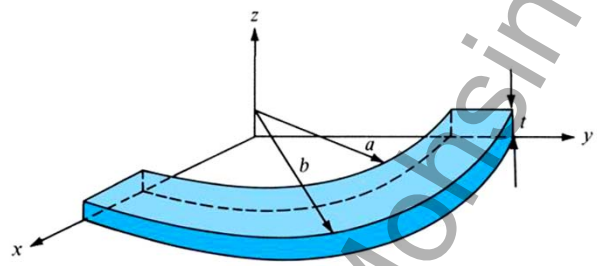
A metal bar of conductivity σ is bent to form a flat 90° sector of inner radius a , outer radius b , and the thickness t as shown in the Figure. Show that:

- a) The resistance of the bar between the vertical curved surfaces at $\rho = a$ and $\rho = b$ is:

$$R = \frac{2 \ln \frac{b}{a}}{\sigma \pi t}$$

- b) The resistance between the two horizontal surfaces at $z = 0$ and $z = t$ is:

$$R' = \frac{4t}{\sigma \pi (b^2 - a^2)}$$



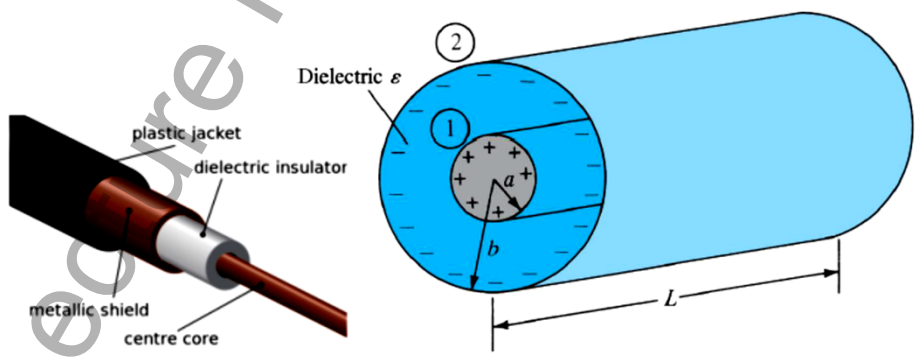
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Practice Example # 02

A coaxial cable contains an insulating material of conductivity σ . If the radius of the central wire is a and that of the sheath is b , show that the conductance of the cable per unit length is:

$$G = \frac{2\pi\sigma}{\ln \frac{b}{a}}$$

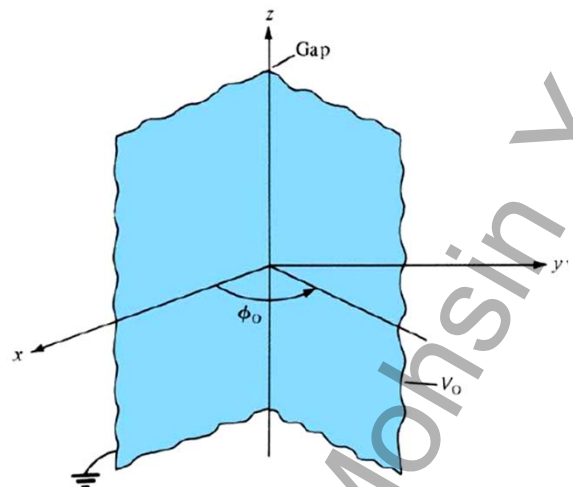


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Practice Example # 03

Semi-infinite conducting planes at $\phi = 0$ and $\phi = \pi/6$ are separated by an infinitesimal insulating gap as shown in Figure. If $V(\phi = 0) = 0$ and $(\phi = \pi/6) = 100 \text{ V}$, Calculate V and $\mathbf{E}$ in the region between the planes.

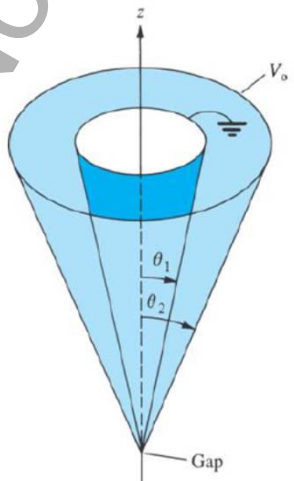


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Practice Example # 04

Two conducting cones ($\theta = \pi/10$ and $\theta = \pi/6$) of infinite extent are separated by an infinitesimal gap at $r = 0$. If $V(\theta = \pi/10) = 0$ and $V(\theta = \pi/6) = 50 \text{ V}$, Find V and $\mathbf{E}$ between the cones.

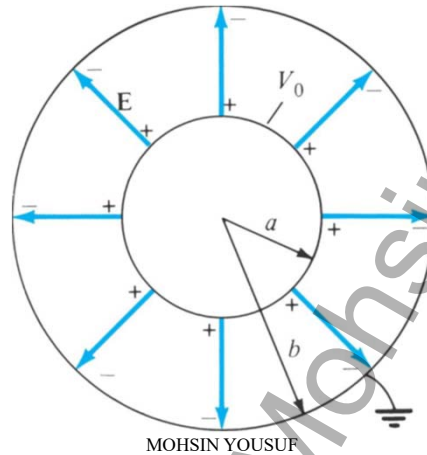


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Practice Example # 05

Conducting spherical shells with radii $a = 10$ cm and $b = 30$ cm are maintained at a potential difference of 100 V such that $V(r = b) = 0$ and $V(r = a) = 100$ V. Determine V and $\mathbf{E}$ in the region between the shells. If $\epsilon_r = 2.5$ in the region, determine the total charge induced on the shells and the capacitance of the capacitor.



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